

Discrete analysis to ROBER PROBLEM

Not assuming QSSA

<1>

$$\begin{bmatrix} [A]_{t+1} \\ [B]_{t+1} \\ [C]_{t+1} \end{bmatrix} = \begin{bmatrix} 1 - k_1 \delta t & 0 & 0 & k_3 \delta t & 0 \\ k_1 \delta t & 1 & 0 & -k_2 \delta t & -k_3 \delta t \\ 0 & 0 & 1 & 0 & k_2 \delta t \end{bmatrix} \begin{bmatrix} [A]_t \\ [B]_t \\ [C]_t \\ [B]_t^2 \\ [B]_t^2 \end{bmatrix}$$

$$\begin{bmatrix} [S]_{t+1} \end{bmatrix} = \underline{\underline{U(S)}} \underline{\underline{\sum_{i=1}^n V_i^T}} \begin{bmatrix} [rec]_t \end{bmatrix}$$

With QSSA assumed on B: $[B]_{t+1} = [B]_t$

<2>

$$\begin{bmatrix} [A]_{t+1} \\ [B]_{t+1} \\ [C]_{t+1} \end{bmatrix} = \begin{bmatrix} 1 - k_1 \delta t & k_3 [B]_t \delta t \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} [A]_t \\ [C]_t \end{bmatrix} + \begin{bmatrix} 0 \\ [B]_t \\ k_2 [B]_t^2 \delta t \end{bmatrix}$$

Set Criteria for QSSA to be assumed:

1. given a species: $T_s = \frac{[S]}{[S]}$ $\frac{\delta t}{T_s} \leq \epsilon \Rightarrow \frac{[S]}{[S]} \delta t \leq \epsilon$

2. for species A : $i \rightarrow$ index for species, $j \rightarrow$ index for time

$$A_{j+1} = A_j + \frac{dA}{dt} \cdot \delta t$$

trying to find out how this 'ε' is going to affect other species.

<3>

$$\begin{aligned}
 &= A_j + \sum_{i=1}^m \frac{\partial A}{\partial S_i} \left(\frac{\partial S_i}{\partial t} \cdot \delta t \right) \rightarrow [S_i] \cdot \delta t \leq \epsilon [S] \\
 &\leq A_j + \sum_{i=1}^m \left(\frac{\partial A}{\partial S_i} \right) \epsilon [S] \rightarrow \text{untraceable analytically}
 \end{aligned}$$

try solving $\frac{dA}{dt}$ using analytical solution:

e.g. ROBER problem: $[A]_t = [A]_0 + \int_0^t -k_1[A] + k_3[B][C] \cdot dt$

$$\frac{\partial [A]}{\partial [B]} = 0 + \frac{\partial}{\partial [B]} \int_0^t -k_1[A] + k_3[B][C] \cdot dt$$



$$\frac{\partial [A]}{\partial [B]} = - \int_0^t \left\{ k_1 \cdot \frac{\partial [A]}{\partial [B]} + k_3 \frac{\partial}{\partial [B]} ([B][C]) \right\} dt$$

untraceable analytically

However, perturbation method can be applied to approximate $\frac{\partial A_i}{\partial A_j}$

Express <3> in a slightly modified way:

$$(A_{t+1} - A_t) = \sum_{j=1}^n \frac{\partial A_i}{\partial A_j} \cdot A_j \cdot \delta_t$$

<4>

$$\begin{bmatrix} A_{t+1} - A_t \end{bmatrix} = \begin{bmatrix} 1 & \frac{\partial A_1}{\partial A_2} & \frac{\partial A_1}{\partial A_3} \\ \vdots & \vdots & \vdots \\ \frac{\partial A_n}{\partial A_1} & \dots & 1 \end{bmatrix}_t \begin{bmatrix} \delta_t \cdot A_1 \\ \delta_t \cdot A_2 \\ \delta_t \cdot A_n \end{bmatrix}$$

in a matrix way.

Reciprocal Matrix: $a_{ij} \cdot a_{ji} = 1$.

<5> proof otherwise ???

ROBER Example:

$$\frac{\partial A}{\partial B} \approx \frac{\Delta A}{\Delta B} = \frac{A(B+\Delta) - A(B)}{\Delta} = \frac{k_3 \Delta \delta_t \cdot C}{\Delta} = k_3 \cdot \delta_t \cdot C$$

$$\frac{\partial C}{\partial B} \approx \frac{\Delta C}{\Delta B} = \frac{K_2(B+\Delta) \delta_t - K_2 B \delta_t}{\Delta} \approx 2K_2 \delta_t B$$

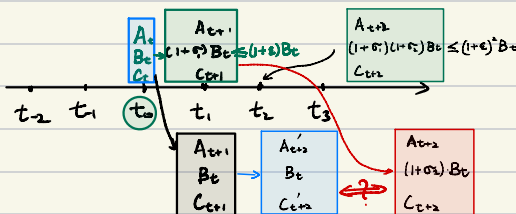
$$\frac{\partial A}{\partial C} \approx k_3 B \cdot \delta_t$$

TRY TO UNDERSTAND HOW 'E' IS GOING TO AFFECT OTHERS?

<fig. 1>

REAL

QSS of B



Trace Species A & C under the QSS assumption:

- Start from $[A_{t+1}, (1+\sigma) B_t, C_{t+1}]^T$
- <2> & <1> give the same updates for A & C
- Compare $[A_{t+2}, B_t, C_{t+2}]^T$ with $[A_{t+2}, (1+\sigma_2) B_t, C_{t+2}]^T$

Calculate $[A_{t+2}, (1+\sigma)B_t, C_{t+2}]^T$:

$$A_{t+2} = (1 - k_1 \delta_t) A_{t+1} + (1+\sigma) k_3 \delta_t B_t \cdot C_{t+1}$$

$$B_{t+2} = (1+\sigma) B_t$$

$$C_{t+2} \approx C_{t+1} + (1+2\sigma) k_2 B_t^2 \delta_t$$

Calculate $[A'_{t+2}, B_t, C'_{t+2}]^T$

$$A'_{t+2} = (1 - k_1 \delta_t) A_{t+1} + k_3 \delta_t B_t \cdot C_{t+1}$$

$$B'_{t+2} = B_t$$

$$C'_{t+2} = C_{t+1} + k_2 B_t^2 \delta_t$$

$$\begin{aligned} \begin{bmatrix} A_{t+2} - A'_{t+2} \\ B_{t+2} - B'_{t+2} \\ C_{t+2} - C'_{t+2} \end{bmatrix} &= \begin{bmatrix} \sigma k_3 \delta_t B_t C_{t+1} \\ \sigma B_t \\ 2\sigma k_2 B_t^2 \delta_t \end{bmatrix} \\ &= \begin{bmatrix} \sigma k_3 B_t \cdot C_{t+1} \\ B_{t+1} \\ 2\sigma k_2 B_t^2 \end{bmatrix} \cdot \delta_t \end{aligned}$$

~

Criteria 2 :

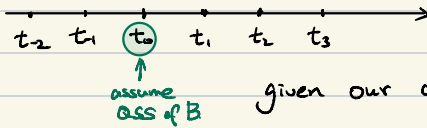
$$\frac{\sigma k_3 B_t \cdot C_{t+1}}{\dot{A}_{t+1}} \leq \varepsilon$$

$$\frac{2\sigma k_2 B_t^2}{\dot{C}_{t+1}} \leq \varepsilon$$

NB: nominator is also a rate, so we do not have to multiply δ_t on it.

TRY TO UNDERSTAND WHEN TO PAUSE THE QSSA

<fig.2>



given our criteria: $\delta_t \leq \varepsilon \cdot \frac{B_t}{B_i}$

$$B \cdot \delta_t = \sigma B_t \quad (\sigma \leq \varepsilon)$$

<6>

Real System:	QSSA System:
$B_{t+1} = (1 + \sigma) B_t$	$B_{t+1} = B_t$

$$\Downarrow$$

$$B_{t+1, \text{QSS}} = \frac{1}{1 + \sigma} B_{t+1, \text{real}}$$

↓ assume this σ is constant (or about the same)
for $T \in \{t_0, t_1, \dots, t_n, \dots\}$

for species B, given ' σ ', and the similarity pre-factor ' ϕ '
closer to one, more similar to the true system value

$$\left(\frac{1}{1 + \sigma}\right)^n = \phi$$

<7>

$$\textcircled{n} = - \frac{\ln \phi}{\ln(1 + \sigma)}$$

of QSSA steps at the acceptable deviation level ' ϕ ' reached

To give a rough idea:

$$\phi = 0.9, \sigma = 1e-4 \rightarrow n \approx 1053 \text{ steps} \quad \phi = 0.99, \sigma = 1e-4 \rightarrow n \approx 100 \text{ steps}$$

$$\phi = 0.9, \sigma = 1e-3 \rightarrow n \approx 105 \text{ steps} \quad \phi = 0.99, \sigma = 1e-3 \rightarrow n \approx 10 \text{ steps}$$

???

CHALLENGE: After n steps, if I relax the QSS assumption, will the system converge to the 'true system' ???

05/04/2015

The 'leaking' in mass seems persist by assuming QSSA:

$$\dot{B} = k_1 A - k_2 B^* - k_3 B C$$

In Sili-Deng / Stiff-PINN-chem-kinetics:

$$0 = k_1 A - k_2 B^* - k_3 B C \Rightarrow B = \frac{-k_3 y_3 + \sqrt{k_3^2 y_3^2 + 4 k_1 k_2 y_1}}{2 k_2}$$

mass not conserved.

To solve this issue, we can apply QSS to find other species' concentration of next step, then we can use molecules' conservation to find the concentration evolution of the QSS species.

from Tianfeng-Lu / 2004-Lu-CNF-DRG: the high reactivity of these radicals renders them ideal QSSA candidates over entire parametric range.

Apr. 15th

from <3> & <4>:

modify <3>, and assume that we have 'm' species, 'n' QSSA-able species, s.t. $n \leq m$:

$$\begin{aligned} S_j^{t+1} &= S_j^t + \frac{\partial S_j^t}{\partial t} \cdot dt = S_j^t + \sum_{i=1}^n \frac{\partial S_j^t}{\partial S_i^{t-1}} \frac{\partial S_i^{t-1}}{\partial t} \cdot dt \\ &= S_j^t + \sum_{i=1}^{m-n} \frac{\partial S_j^t}{\partial S_i^{t-1}} \frac{\partial S_i^{t-1}}{\partial t} \cdot dt + \sum_{k=1}^n \frac{\partial S_j^t}{\partial S_k^{t-1}} \boxed{\frac{\partial S_k^{t-1}}{\partial t} \cdot dt} \\ &\leq \varepsilon_k \cdot S_k^{t-1} \end{aligned}$$

the influence can only appear from last time step, not this time step.

<4> can be re-written as:

$$\begin{bmatrix} S_1^{t+1} - S_1^t \\ \vdots \\ S_{m-n}^{t+1} - S_{m-n}^t \\ S_{m-n+1}^{t+1} - S_{m-n+1}^t \\ \vdots \\ S_m^{t+1} - S_m^t \end{bmatrix} = \begin{bmatrix} \frac{\partial S_1^t}{\partial S_1^{t-1}} & \frac{\partial S_1^t}{\partial S_2^{t-1}} & \dots & \frac{\partial S_1^t}{\partial S_{m-n}^{t-1}} & \frac{\partial S_1^t}{\partial S_{m-n+1}^{t-1}} & \dots & \frac{\partial S_1^t}{\partial S_m^{t-1}} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{\partial S_{m-n}^t}{\partial S_1^{t-1}} & \frac{\partial S_{m-n}^t}{\partial S_2^{t-1}} & \dots & \frac{\partial S_{m-n}^t}{\partial S_{m-n}^{t-1}} & \frac{\partial S_{m-n}^t}{\partial S_{m-n+1}^{t-1}} & \dots & \frac{\partial S_{m-n}^t}{\partial S_m^{t-1}} \\ \frac{\partial S_{m-n+1}^t}{\partial S_1^{t-1}} & \frac{\partial S_{m-n+1}^t}{\partial S_2^{t-1}} & \dots & \frac{\partial S_{m-n+1}^t}{\partial S_{m-n}^{t-1}} & \frac{\partial S_{m-n+1}^t}{\partial S_{m-n+1}^{t-1}} & \dots & \frac{\partial S_{m-n+1}^t}{\partial S_m^{t-1}} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{\partial S_m^t}{\partial S_1^{t-1}} & \frac{\partial S_m^t}{\partial S_2^{t-1}} & \dots & \frac{\partial S_m^t}{\partial S_{m-n}^{t-1}} & \frac{\partial S_m^t}{\partial S_{m-n+1}^{t-1}} & \dots & \frac{\partial S_m^t}{\partial S_m^{t-1}} \end{bmatrix} \begin{bmatrix} \frac{\partial S_1^{t-1}}{\partial t} \cdot dt \\ \frac{\partial S_2^{t-1}}{\partial t} \cdot dt \\ \vdots \\ \frac{\partial S_{m-n}^{t-1}}{\partial t} \cdot dt \\ \frac{\partial S_{m-n+1}^{t-1}}{\partial t} \cdot dt \\ \vdots \\ \frac{\partial S_m^{t-1}}{\partial t} \cdot dt \end{bmatrix}$$

not going to be affected

(I) (II) (III) (IV)

$\leq \varepsilon_k \cdot S_k^{t-1}$

$$\Delta S_i = S_i^{t+1} - S_i^t = f\left(\frac{\partial S_i^t}{\partial S_j^t} \cdot \frac{\partial S_j^t}{\partial t} \cdot dt\right)$$

- ①: described by eq-8
 ②: NOT AFFECT immediately need to be 'small' s.t. QSSA works
 ③: ? in QSSA, they will be zero
 ④: ? in QSSA, they will be zero

Jun. 13th

Derivation of policy 1 & policy 2:

POLICY 1: species that assumed to be QSS in n step interval should have relative error no more than " ϵ "

from <6> to <9> $\begin{cases} B_{t+n} = (1 + \sigma) B_t, \text{ where } \sigma \text{ is a small number} \\ B_{QSSA, t+n} = B_t \end{cases}$

assume for next n steps,
 $t \in [1, n], B_{t+1} = (1 + \Delta) B_t, \Delta \leq \sigma$

Stringent Requirement

RELAX

assume for next n steps,
 $t \in [1, n], B_{t+1} = (1 + \Delta) B_t,$
 $\Delta \neq \sigma$

relaxed Requirement

Test after n steps:

if $\frac{n \cdot \text{pr}_{min} \cdot \sigma_{max}}{B_{tn}} \gg \sigma$, GO BACK

derive to

$$\frac{B_{t+n+1}}{B_{t+n}} \gg 1 + \sigma$$

ensure that QSSA does not cause LARGE change
 in B 's concentration after n -steps:
 (<9>)

from <9> to <10>

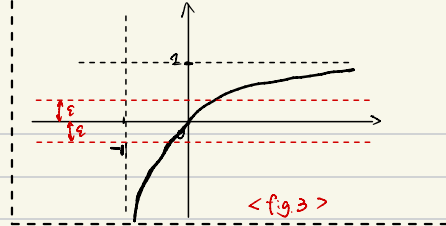
$$\frac{B_{t+n} - B_{QSSA}}{B_{tn}} \leq \frac{[(1 + \sigma)^n - 1] \cdot B_{QSSA}}{(1 + \sigma)^n \cdot B_{QSSA}}$$

expand, ignore higher order terms

$$\approx \frac{n\sigma}{1 + n\sigma}$$

re-write (10):

$$\frac{B_{t+n} - B_{ssa}}{B_{t+n}} \leq \frac{n\sigma}{1+n\sigma}$$



recall from policy 1, the abs. rel. error, after n -steps of the QSSA assumption, would be no larger than ' ϵ '.

<11>: $\left| \frac{n\sigma}{1+n\sigma} \right| \leq \epsilon$, we know $\epsilon \ll 1$, $n\sigma > -1$ *

$\sigma > 0$: $\frac{n\sigma}{1+n\sigma} \leq \epsilon$

$\sigma \leq 0$: $-\frac{n\sigma}{1+n\sigma} \leq \epsilon$

$\sigma > 0$: $n\sigma \leq \frac{\epsilon}{1-\epsilon}$

$\sigma \leq 0$: $n\sigma \geq -\frac{\epsilon}{1-\epsilon}$

$-\frac{\epsilon}{1+\epsilon} \leq n\sigma \leq \frac{\epsilon}{1-\epsilon}$

<12>

$$-\frac{\epsilon}{(1+\epsilon) \cdot n} \leq \sigma \leq \frac{\epsilon}{(1-\epsilon) \cdot n}$$

effectively, what is σ ? n.p.r

$$\sigma = \frac{w \cdot \Delta t}{B_t} \leftarrow \begin{cases} B_{t+1} = B_t + w \cdot \Delta t \\ B_{t+1} = (1+\sigma) B_t \end{cases}$$



<13>

$$\sigma = u \cdot \Delta t$$

POLICY 2: By assuming QSS for some species, other species' concentration will not be affected too much, s.t. the abs. rel. error is more than ' ϵ '

<14>

$$S_i^{t+1} - S_i^t = \underbrace{\int_{\frac{t}{\Delta t}}^{\frac{t+1}{\Delta t}} \frac{\partial S_i^t}{\partial S_j^{t+1}} \cdot \dot{S}_j^{t+1} dt}_{\Delta - \delta} + \underbrace{\int_{\frac{t}{\Delta t}}^{\frac{t+1}{\Delta t}} \frac{\partial S_i^t}{\partial S_j^t} \cdot \dot{S}_j^{t+1} dt}_{\delta}$$

$$S_i^{t+1} = S_i^t + (\Delta - \delta) \cdot dt + \delta \cdot dt$$

$$\text{QSSA } \tilde{S}_i^{t+1} = S_i^t + (\Delta - \delta) \cdot dt$$

one-step-error: $\text{error}_{1\text{-step}} = \delta \cdot dt$

ignoring higher order effect, after n -steps:

$$S_i^{t+n} - \tilde{S}_i^{t+n} = \delta \cdot n \cdot dt$$

recall from policy 2, we require:

<15>

$$\frac{S_i^{t+n} - \tilde{S}_i^{t+n}}{S_i^{t+n}} = \frac{\delta \cdot n \cdot dt}{\tilde{S}_i^{t+n} + \delta \cdot n \cdot dt} = \frac{\delta \cdot n \cdot dt}{S_i^t + \sum_{j=1}^n (\Delta - \delta) \cdot dt_j + \frac{\delta}{j+1} \cdot \delta \cdot dt_j}$$

CANNOT EASILY ASSESSED

$\approx$

$$\frac{\delta \cdot n \cdot dt}{S_i^t + \sum_{j=1}^n (\Delta - \delta) \cdot dt_j + \delta \cdot dt \cdot n}$$

variation of this between time steps are not negligible!

another method to do it:

$$\frac{\delta}{\Delta} < \epsilon, \text{ } \epsilon \text{ requires try and error}$$

Numerical Challenges in this process:

In policy 1:

when density becomes extremely small, e.g., $\sigma = \frac{\omega \cdot dt}{B_t} = \mu \cdot \Delta t$

has questionable accuracy.

Solution: Species that has concentration lower than 10^{-16} should be considered **PASSING POLICY I**

In policy 2: $\frac{\partial S_i^t}{\partial S_j^{t-1}} \cdot \dot{S}_j^{t-1} \Delta t$ can be challenging to access

when $\dot{S}_j^{t-1}$ is very small

<16>

$$\text{Solution: } \frac{\partial S_i^t}{\partial S_j^{t-1}} \cdot \dot{S}_j^{t-1} \Delta t \approx \frac{\tilde{S}_i^t - S_i^t}{\dot{S}_j^{t-1} \cdot \Delta t - 0} \cdot \dot{S}_j^{t-1} \Delta t$$



$$\frac{\partial S_i^t}{\partial S_j^{t-1}} \cdot \dot{S}_j^{t-1} \Delta t \approx \tilde{S}_i^t - S_i^t \Rightarrow$$

for species that has concentration non-zero, we can also use method similar to <eq. 16>

$$\frac{\partial S_i^t}{\partial S_j} \cdot \dot{S}_j^{t-1} \Delta t = \frac{\tilde{S}_i^t - S_i^t}{\dot{S}_j^{t-1} \Delta t} \cdot \dot{S}_j^{t-1} \Delta t = \tilde{S}_i^t - S_i^t$$

Thoughts on perturbing S or P ?

constant volume $\rightarrow$ perturbing P (mole composition based)

constant pressure $\rightarrow$ perturbing S . (mass composition based)